

A Hilbert-Operator Partial Result for Hanson Question 4.3

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Source Question

James Hanson asks in *Indiscernible Subspaces and Minimal Wide Types*, arXiv:2004.03062, Question 4.3:

Does (model theoretic) stability imply oscillation stability? That is to say, if T is a stable Banach theory, is every unary formula oscillation stable on models of T ?

The arXiv record notes that this preprint was absorbed into arXiv:2011.00610, *Strongly Minimal Sets and Categoricity in Continuous Logic*; the corrected absorbing paper keeps the same question in its Banach-theory section.

Result

This note proves a positive subcase for quantifier-free formulas in Hilbert structures expanded by bounded linear operators in the standard many-sorted operator-linear Hilbert/Banach language. The result does not use model-theoretic stability. Its relevance is that many natural examples near Hanson's question are stable Hilbert structures with distinguished linear operators and quantifier elimination.

Proof Intuition

A unary quantifier-free Hilbert-operator formula only uses finitely many atoms of the form $\|Sx+b\|$, where S is a bounded operator obtained from the named operators by finitely many algebraic operations and b is a parameter vector. On an infinite-dimensional subspace, the finitely many linear terms $\langle Sx, b \rangle$ can be killed by passing to a finite-codimensional subspace. What remains is a finite list of quadratic forms $\langle S^*Sx, x \rangle$. A diagonal orthonormal-subsequence argument produces an infinite-dimensional subspace on which all those quadratic forms are simultaneously almost constant. Uniform continuity of the continuous connectives then gives small oscillation for the whole formula.

A Compression Lemma

Lemma 1 (simultaneous quadratic flattening). *Let H be an infinite-dimensional real or complex Hilbert space, let $A_1, \dots, A_m \in B(H)$ be bounded self-adjoint operators, and let $\eta > 0$. Then there are an infinite-dimensional closed subspace $Z \subset H$ and scalars $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that, for every unit vector $z \in Z$,*

$$|\langle A_j z, z \rangle - \lambda_j| < \eta \quad (1 \leq j \leq m).$$

Proof. Choose an orthonormal sequence (u_n) in H . By compactness of $\prod_{j=1}^m [-\|A_j\|, \|A_j\|]$, after passing to a subsequence we may assume that $\langle A_j u_n, u_n \rangle \rightarrow \lambda_j$ for every j .

We now pass to a further subsequence (e_n) recursively. Since every orthonormal sequence is weakly null, once $e_1, \dots, e_{n-1}$ have been chosen, a sufficiently far tail vector u_k satisfies

$$|\langle A_j u_k, u_k \rangle - \lambda_j| < \eta/4$$

for all j , and also

$$|\langle A_j u_k, e_r \rangle| < \frac{\eta}{16m} 2^{-n-r} \quad (1 \leq j \leq m, r < n).$$

Relabel the resulting subsequence as (e_n) , and let $Z = \overline{\text{span}}\{e_n : n \geq 1\}$.

Fix j and write a unit vector $z \in Z$ as $z = \sum_n c_n e_n$. Then

$$\begin{aligned} |\langle A_j z, z \rangle - \lambda_j| &\leq \sum_n |c_n|^2 |\langle A_j e_n, e_n \rangle - \lambda_j| \\ &\quad + \sum_{n \neq r} |c_n| |c_r| |\langle A_j e_n, e_r \rangle|. \end{aligned}$$

The diagonal part is $< \eta/4$. For the off-diagonal part, use $(\sum_n 2^{-n} |c_n|)^2 \leq \sum_n 4^{-n} < 1$. The recursive bounds, together with self-adjointness for the reversed pairs, make this part $< \eta/(16m)$. Hence the whole expression is $< \eta$. $\square$

The Partial Theorem

Theorem 1. *Let H be an infinite-dimensional Hilbert space in the standard many-sorted operator-linear Hilbert/Banach language, expanded by any collection of bounded linear operator symbols. Let $\varphi(x, \bar{a})$ be a quantifier-free unary formula with parameters $\bar{a}$, whose one-variable atoms are norms of operator-linear terms. For every infinite-dimensional closed subspace $Y \subseteq H$ and every $\varepsilon > 0$, there exists an infinite-dimensional closed subspace $Z \subseteq Y$ such that*

$$\sup\{|\varphi(u, \bar{a}) - \varphi(v, \bar{a})| : u, v \in S_Z\} < \varepsilon.$$

Proof. Only finitely many atomic predicates occur in φ . By the operator-linear language hypothesis, each unary term is of the form $Tx + c$, where T is a bounded operator obtained as a finite linear combination of finite words in the named operator symbols, and c is a parameter vector. Hence each unary atomic predicate appearing in φ has the form

$$x \mapsto \|Sx + b\|$$

for some bounded operator S and vector b .

Let $\|S_1 x + b_1\|, \dots, \|S_N x + b_N\|$ list these atoms. By uniform continuity of the continuous connective defining φ , choose $\delta > 0$ so small that changing every atomic value by at most δ changes φ by less than ε .

Inside Y , remove the finitely many linear terms by setting

$$Y_0 = Y \cap \bigcap_{\ell=1}^N \ker(z \mapsto \langle S_\ell z, b_\ell \rangle).$$

This is still infinite-dimensional. For $z \in Y_0$ with $\|z\| = 1$,

$$\|S_\ell z + b_\ell\|^2 = \langle S_\ell^* S_\ell z, z \rangle + \|b_\ell\|^2.$$

Apply Lemma 1 in Y_0 to the finite self-adjoint family $S_\ell^* S_\ell$ restricted to Y_0 , with tolerance chosen small enough that the corresponding square-rooted norms vary by at most δ . The result is an infinite-dimensional closed subspace $Z \subseteq Y_0$ on which every atom $\|S_\ell x + b_\ell\|$ varies by at most δ over S_Z . Therefore $\varphi(x, \bar{a})$ varies by less than ε over S_Z , as required. $\square$

Transfer to Quantifier-Elimination Classes

Corollary 1 (QE transfer). *Let T be a complete theory of infinite-dimensional Hilbert spaces in a language obtained by adding bounded linear operator symbols to the standard Hilbert/Banach language. Suppose that every unary formula of T is, over T , uniformly approximable by quantifier-free unary formulas whose atoms are norms of operator-linear terms. Then every unary formula of T is oscillation stable on every model of T .*

Proof. Let $\varphi(x, \bar{a})$ be a unary formula, Y an infinite-dimensional closed subspace of a model of T , and $\varepsilon > 0$. Choose a quantifier-free $\psi(x, \bar{a})$ of the kind handled in Theorem 1 such that $|\varphi - \psi| < \varepsilon/3$ on the relevant sort. Theorem 1 gives an infinite-dimensional closed $Z \subseteq Y$ on whose unit sphere ψ has oscillation $< \varepsilon/3$. Then φ has oscillation $< \varepsilon$ on S_Z . $\square$

Corollary 2 (named stable Hilbert-operator classes). *Hanson’s Question 4.3 has a positive answer for the following stable Hilbert-operator theories.*

1. *Completions of the theory of Hilbert spaces expanded by a bounded normal operator T , in the language with T and T^* .*
2. *The self-adjoint and unitary one-operator subcases in the original language with just the named operator.*
3. *Hilbert spaces expanded by a unitary representation of a finite group.*

Proof. For (1), Berenstein–Cuervo Ovalle–Goldbring prove quantifier elimination for the completions after adding T^* (Theorem 4.1 of arXiv:2507.21894) and prove that the completions are stable, indeed superstable (Theorems 6.2 and 6.3 of arXiv:2507.21894). Their quantifier-free terms are built from T, T^* , hence are operator-linear terms covered by Corollary 1.

The same paper proves quantifier elimination in the original language for the self-adjoint and unitary cases (Theorem 4.3 of arXiv:2507.21894), so (2) follows in the same way. For (3), Berenstein–Perez prove quantifier elimination for Hilbert spaces expanded by a unitary representation of a finite group (Proposition 3.6 of arXiv:2409.03923) and $\aleph_0$ -stability (Theorem 3.8 of arXiv:2409.03923). Again the terms are finite linear combinations of named unitary actions, so Corollary 1 applies. $\square$

Limitations

Theorem 1 is not a full answer to Hanson Question 4.3. It applies to quantifier-free unary formulas in the standard operator-linear Hilbert/Banach language. Corollary 2 gives full positive answers for several named stable Hilbert-operator theories, but it still does not cover arbitrary stable Banach theories or stable expansions by general unary predicates.

Novelty Check

The run indexes were searched for arXiv:2004.03062, *Indiscernible Subspaces*, minimal wide types, and oscillation stability. Web searches for the exact phrase “Does model theoretic stability imply oscillation stability” and close variants found the source and the absorbing paper, but no explicit later answer. Further searches found the normal-operator and finite-group Hilbert expansion papers cited below; these provide named stable QE classes to which the transfer corollary applies.

References

- James Hanson, *Indiscernible Subspaces and Minimal Wide Types*, arXiv:2004.03062.
- James Hanson, *Strongly Minimal Sets and Categoricity in Continuous Logic*, arXiv:2011.00610.
- Alexander Berenstein, Nicolás Cuervo Ovalle, and Isaac Goldbring, *Model theory of Hilbert spaces expanded by normal operators*, arXiv:2507.21894.
- Alexander Berenstein and Juan Manuel Perez, *Model Theory of Hilbert Spaces with a Discrete Group Action*, arXiv:2409.03923.